Supplemental Document

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Energy yield modelling of textured perovskite/silicon tandem photovoltaics with thick perovskite top cells: supplement

FABRIZIO GOTA,[1,2] RAPHAEL SCHMAGER,[1,2] AHMED FARAG,[1,2] AND ULRICH W. PAETZOLD[1,2,*]

[1] *Institute of Microstructure Technology, Karlsruhe Institute of Technology, Hermann-von-Helmholtz-Platz 1, 76344 Eggenstein-Leopoldshafen, Germany*
[2] *Light Technology Institute, Karlsruhe Institute of Technology, Engesserstrasse 13, 76131 Karlsruhe, Germany*
[*] *ulrich.paetzold@kit.edu*

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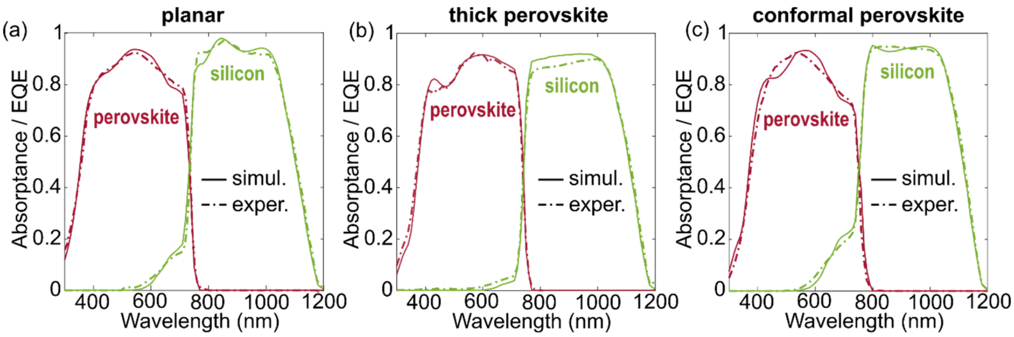

Fig. S1. EQE of experimental cells and simulated absorptance for three perovskite/silicon 2T tandem solar cells with different morphology configurations. For each experimental cell, the layer stack reported in the corresponding publication was used to compute the simulated absorptance. The planar configuration EQE data (a) are extracted from Al-Ashouri et al.[1], the thick perovskite configuration EQE data from De Bastiani et al.[2] and the conformal perovskite configuration EQE data from Aydin et al.[3].

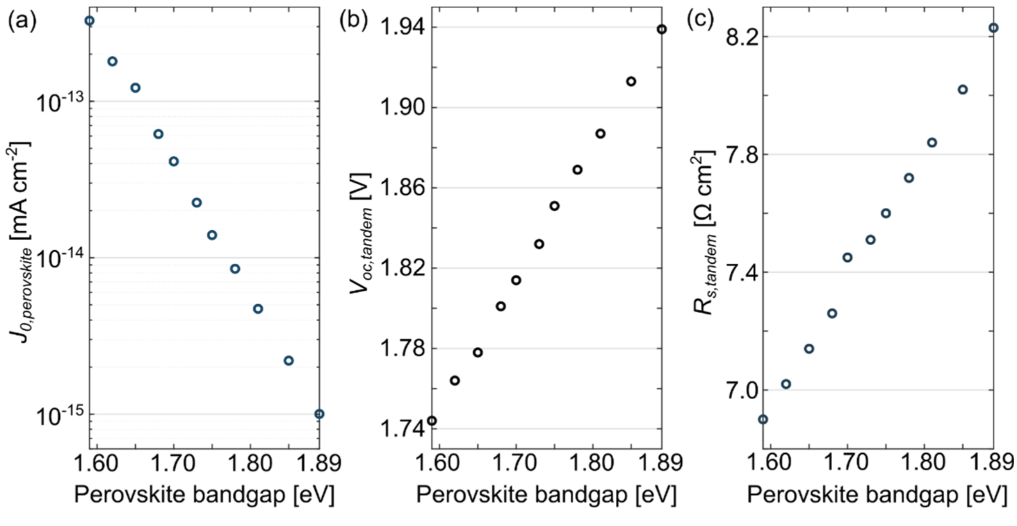

Fig. S2. Electrical parameters used for the simulations in this work. The dark saturation current density of the perovskite absorber $J_{0,perovskite}$ (a) was calculated so that the ratio between the tandem open circuit voltage $V_{oc,tandem}$ and ($E_{g,silicon}$ + $E_{g,perovskite}$)/q is equal to ~65%. In this way, $V_{oc,tandem}$ has a linear dependence on the perovskite bandgap (b). The series resistance of the tandem cells $R_{s,tandem}$ (c) is selected so that the FF of the cell under STCs is equal to 80%. The ideality factor of the perovskite absorber is always set to 1.3. For what concerns the silicon bottom cell, its dark saturation current $J_{0,silicon}$ is set to $3 \cdot 10^{-11}$ mA cm^{-2} and the ideality factor to 0.96. Finally, the shunt resistance of the perovskite top cell is set to 1200 Ω cm^2, while for the silicon bottom cell to 1500 Ω cm^2.

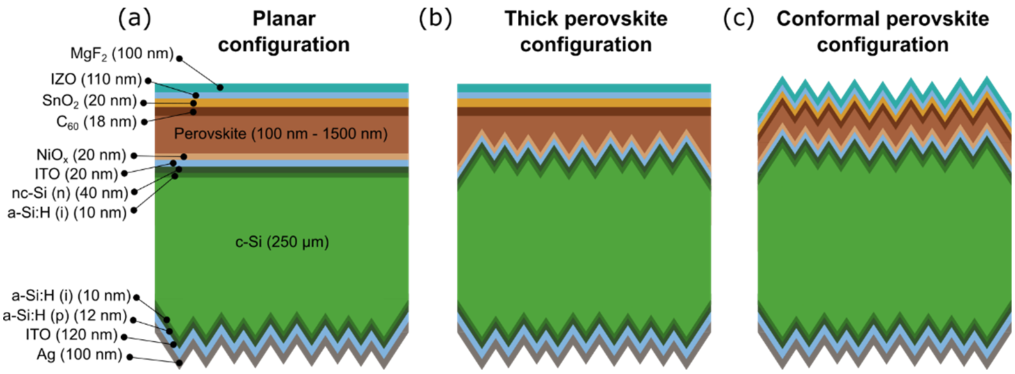

Fig. S3. Schematic cross-section of the layer stacks used in this study for non-encapsulated perovskite/silicon tandem solar cells with planar (a), thick perovskite (b) and conformal perovskite (c) morphologies.

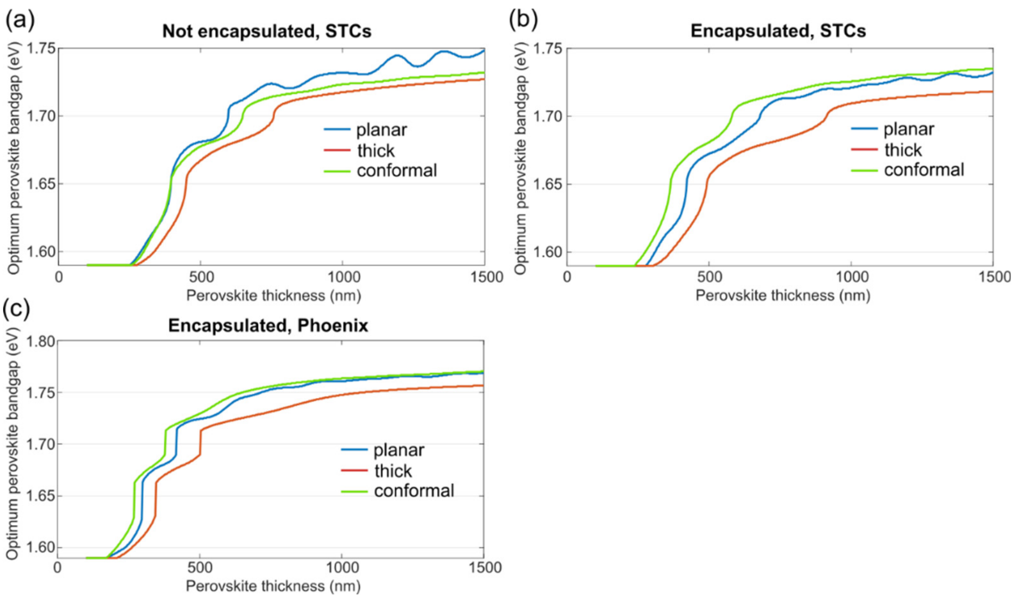

Fig. S4. Optimum perovskite bandgap versus perovskite thickness for non-encapsulated cells in STCs (a), encapsulated cells in STCs (b) and under realistic irradiation conditions in Phoenix, Arizona (c).

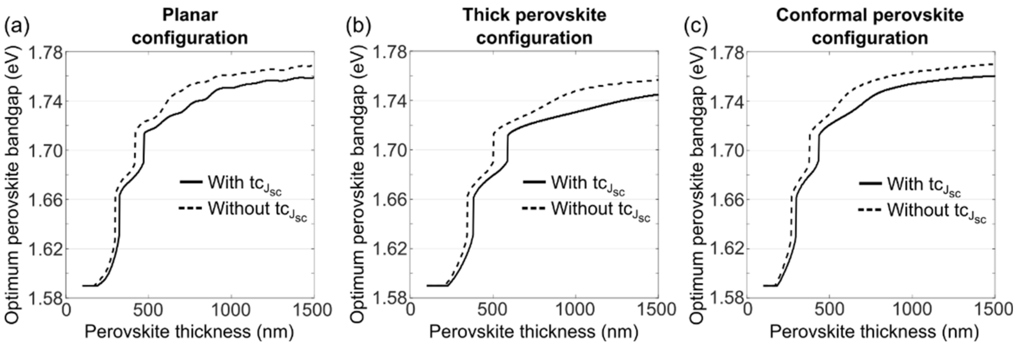

Fig. S5. Optimum perovskite bandgap versus perovskite thickness for encapsulated cells under realistic conditions in Phoenix, Arizona with and without the temperature coefficient for short-circuit current density (tc_{Jsc}) for planar (a), thick perovskite (b) and conformal perovskite (c) morphology configurations. The temperature coefficients for the perovskite top cell and silicon bottom cell short-circuit current density have been extracted from Aydin et al.[3]. With tc_{Jsc}, the optimum perovskite bandgap shifts to lower values, on average of ~15meV.

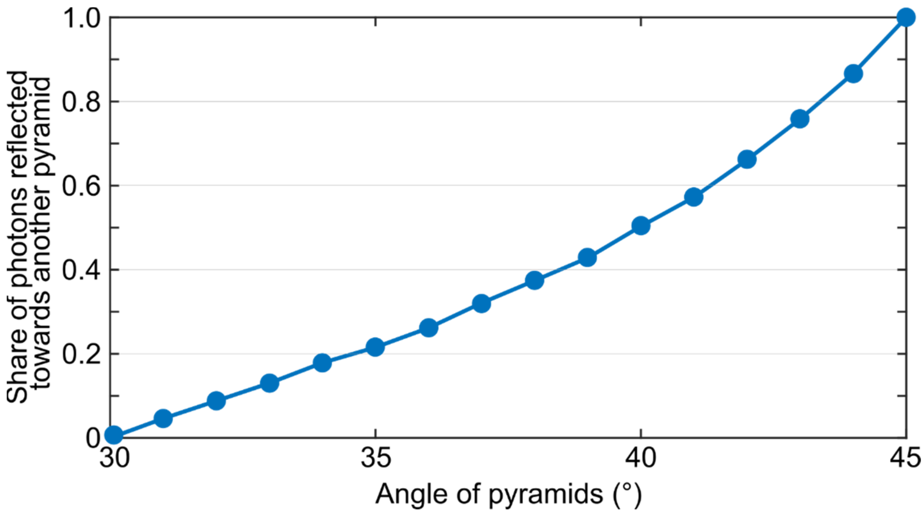

Fig. S6. Share of photons for different characteristic angle of the pyramids that, after having been reflected at a pyramid (with random distribution) interface for normal angle of incidence, impinge again on another pyramid surface. Data are calculated via OPAL 2[4]. For angles ≤ 30°, all photons are directly reflected back to the starting medium of propagation. For angles ≥ 45°, all photons, after having been reflected at the pyramids interface, impinge at least once more on another pyramid interface. Therefore, reflection at this interface is greatly reduced.

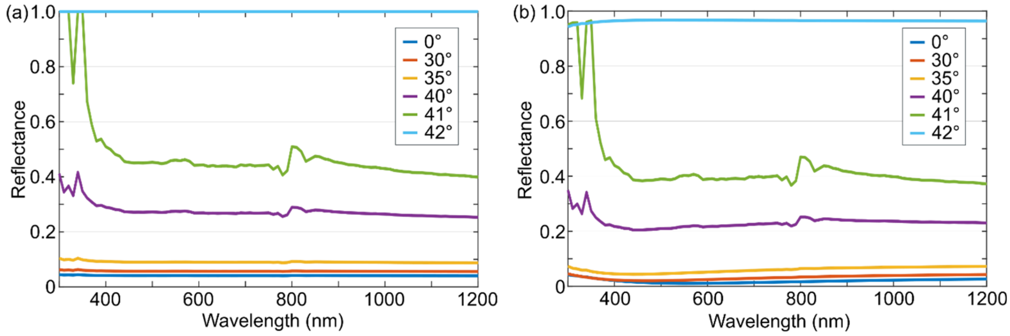

Fig. S7. Reflectance at the glass/air (a) and glass/MgF_2/air (b) interface for different angles of incidence (for light propagating in the glass medium). Without MgF_2, total internal reflection takes places with an angle of incidence of 42° or larger. With MgF_2, for an angle of incidence ≥ 42°, part of the light out-couples from the cell, but reflectance is still very high. For an angle of 42°, the average reflectance is > 0.96%.

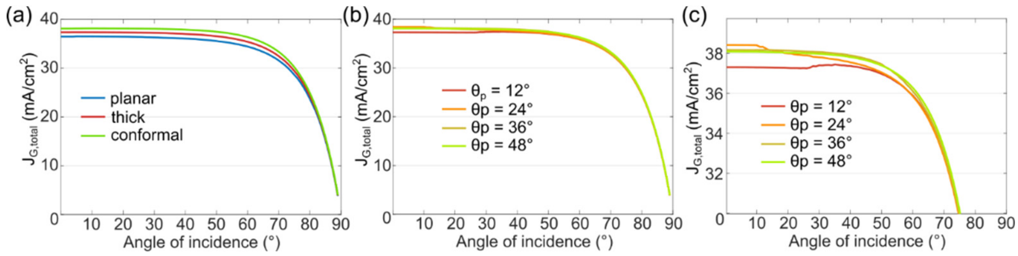

Fig. S8. Total photogenerated current density ($J_{G,total}$) under AM1.5G spectrum as a function of the angle of incidence (AOI) for the three morphology configurations studied in the manuscript (a) and for fully textured c-Si cells with intermediate perovskite front-side angles θ_p (b). (c) Zoom of graph (b) for $J_{G,total} > 30$ mA/cm^2. For normal light incidence, $\theta_p = 24°$ is the optimum angle. However, at large angles of incidence, higher values of θ_p induce a larger total photogenerated current.

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